

HOMS

Practical Activity

Motion in One Dimension

Physical Sciences | Grade 10 | Term 3

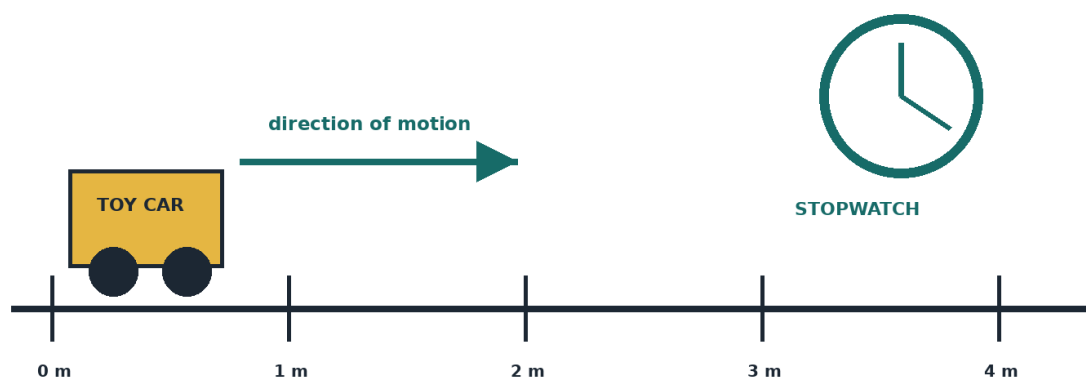
Measure a moving object's position and use a graph to determine velocity.

Learner name	
Date	

CAPS-aligned draft. Educator or subject-expert approval is required before classroom use.

PRACTICAL: MEASURING MOTION

Use a toy car, straight track, metre rule and stopwatch



Measure position at equal time intervals. Plot position against time. Use the gradient to determine average velocity.

Investigation Question

How does the position of a toy car change with time as it moves along a straight track?

Materials

- Toy car or trolley
- Straight track or smooth floor
- Metre rule or measuring tape
- Masking tape or chalk markers
- Stopwatch or phone timer
- Graph paper

Variables

Independent variable	Time
Dependent variable	Position relative to the origin
Controlled variable	Same car, track and release method
Positive direction	From the origin toward the end of the track

Method

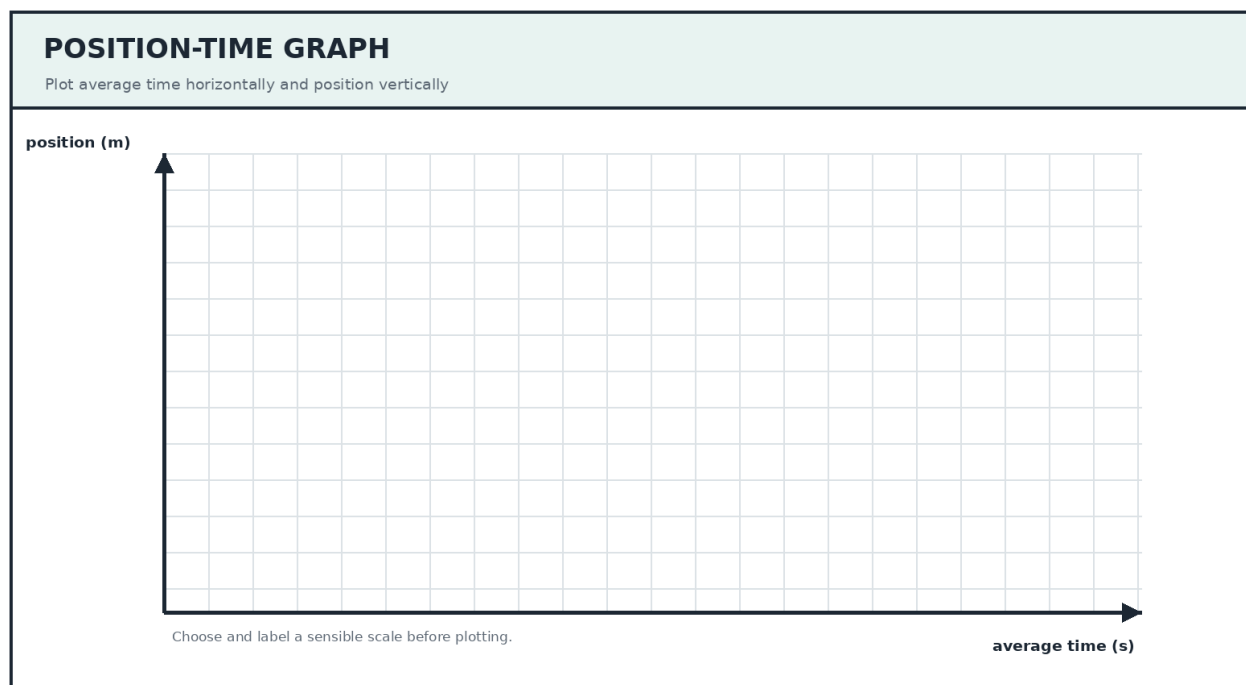
- Mark an origin and positions at 0.50 m intervals along a straight track.
- Place the car at the origin and practise releasing it without pushing.
- Release the car and record the time when its front reaches each position marker.
- Repeat the run three times. Use the average time for each position.
- Plot position on the vertical axis and average time on the horizontal axis.
- Draw a best-fit line or curve and calculate the gradient over a suitable interval.

Safety: Keep the track clear, do not place the car near stairs, and assign one learner to time while another observes position.

Results

Position (m)	Run 1 (s)	Run 2 (s)	Run 3 (s)	Average time (s)	Interval velocity (m.s ⁻¹)
0.0					
0.5					
1.0					
1.5					
2.0					
2.5					

Plot your position-time graph on the grid below. Label both axes and show the points used for one gradient calculation.



Analysis And Evaluation

1 State the relationship between position and time shown by your graph. **(2)**

2 Calculate the average velocity over one suitable interval. Show the points used. **(4)**

3 Was the velocity constant? Use the graph as evidence. **(3)**

4 Identify one source of random error and explain how repeating the run reduces its effect. **(3)**

5 Suggest one improvement that would make the timing more reliable. **(2)**

6 Write a conclusion that answers the investigation question. **(3)**
